# Gate 08 — Husbandry & Breeding Guidance: Backend

## Goal

Ship the data model, Django admin authoring UX, and read-only public API for per-species husbandry & breeding guidance, following the `ConservationAssessment` governance pattern (human-reviewed, attributed, sourced). After this gate lands, Aleksei can author the 3–5 exemplar records from `data/husbandry/TEMPLATE.md` in Django admin before the ECA Workshop, and the frontend (Gate 09) has a stable contract to render against.

No UI work lives in this gate. The frontend teaser + `/species/[id]/husbandry/` route are Gate 09.

## Stories

- **Story 08.1** — As an admin, I want a `SpeciesHusbandry` model tied one-to-one to `Species` with structured fields for water, tank, diet, behavior, breeding, difficulty factors, sourcing, sources, and governance, so that I can author husbandry content that matches `data/husbandry/TEMPLATE.md` without lossy free-text dumps.
- **Story 08.2** — As an admin, I want difficulty surfaced as **factors** (not a single label), so that the page describes *why* a species is challenging rather than issuing a prescriptive verdict (locked decision Q2, 2026-04-18).
- **Story 08.3** — As an admin, I want Django admin to refuse publishing a `SpeciesHusbandry` record unless at least one `HusbandrySource` citation is attached and `last_reviewed_by` + `last_reviewed_at` are populated, so that every public husbandry page is traceable.
- **Story 08.4** — As an admin, I want `published` to be an explicit boolean independent of record existence, so that I can draft records in admin without exposing them on the public site until ready.
- **Story 08.5** — As a Tier 1 public consumer of the REST API, I want to retrieve a species' published husbandry guidance at a stable endpoint, so that the Next.js frontend can render `/species/[id]/husbandry/`.
- **Story 08.6** — As a Tier 1 consumer of the species detail API, I want a `husbandry.has_published_record` boolean (and nothing else) embedded on the existing `/api/species/{id}/` response, so that the profile page can decide whether to render the teaser block without a second request (AC-1 / AC-2).
- **Story 08.7** — As an admin editing a record whose `last_reviewed_at` is older than 24 months, I want the admin form to show a "review pending" indicator, so that I can see stale records at a glance (mirrors the public AC-4 rendering).

## Scope Assessment

| Story | Frontend | Backend | Full-Stack | Complexity |
|---|---|---|---|---|
| 08.1 | | ✓ | | M |
| 08.2 | | ✓ | | S |
| 08.3 | | ✓ | | S |
| 08.4 | | ✓ | | S |
| 08.5 | | ✓ | | M |
| 08.6 | | ✓ | | S |

| | | | | |
|---|---|---|---|---|
| 08.7 | | ✓ | | S |

Total: backend-only gate. No frontend changes.

## Data Model

New Django app `husbandry`. **Confirmed 2026-04-18** by architecture + BA review (see "App location" under Risks and Open Questions). Rationale: matches the project's one-bounded-context-per-app pattern ( `populations` , `fieldwork` , `conservation` ); keeps husbandry migrations off the hot `species` table; gives Tier 3+ submissions / photos / version history a clean home.

### `SpeciesHusbandry`

One-to-one with `Species`. Nullable — absence is the default state. All value fields are nullable/blank; the template drives which are populated.

- `species` — `OneToOneField(Species, on_delete=CASCADE, related_name='husbandry')` .
- `published` — `BooleanField(default=False)` . Gates public API + frontend teaser.
- **Water** — `water_temp_c_min` / `water_temp_c_max` (Decimal, nullable), `water_ph_min` / `water_ph_max` (Decimal, nullable), `water_hardness_dgh_min` / `_dgh_max` / `_dkh_min` / `_dkh_max` (Decimal, nullable), `water_flow` (CharField with choices `still` / `gentle` / `moderate` / `strong` , blank), `water_notes` (TextField, blank).
- **Tank** — `tank_min_volume_liters` (PositiveInteger, nullable), `tank_min_footprint_cm` (CharField, blank — free text like "120x45"), `tank_aquascape` / `tank_substrate` / `tank_cover` / `tank_notes` (all TextField, blank).
- **Diet** — `diet_accepted_foods` (JSONField `list` , default=list), `diet_live_food_required` (Boolean, default False), `diet_feeding_frequency` (CharField, blank), `diet_notes` (TextField, blank).
- **Behavior** — `behavior_temperament` / `behavior_recommended_sex_ratio` / `behavior_schooling` / `behavior_community_compatibility` / `behavior_notes` (all CharField or TextField, blank).
- **Breeding** — `breeding_spawning_mode` (CharField with choices `substrate_spawner` / `mouthbrooder` / `annual_killi` / `bubble_nest` / `livebearer` / `other` , blank), `breeding_triggers` / `breeding_egg_count_typical` / `breeding_fry_care` / `breeding_survival_bottlenecks` / `breeding_notes` (TextField, blank).
- **Difficulty factors** (locked Q2: factors, not a single label) — `difficulty_adult_size` , `difficulty_space_demand` , `difficulty_temperament_challenge` , `difficulty_water_parameter_demand` , `difficulty_dietary_specialization` , `difficulty_breeding_complexity` , `difficulty_other` — all CharField, blank. **No aggregate `difficulty` column.**
- **Sourcing** — `sourcing_cares_registered_breeders` (Boolean, default False), `sourcing_notes` (TextField, blank). The narrative ethics block itself is rendered by the existing `HusbandrySourcingEthics.tsx` component and is not stored per-record.
- **Narrative** — `narrative` (TextField, blank — Markdown, 2–5 short paragraphs per template guidance).
- **Governance** — `contributors` (TextField, blank — free text at MVP per BA §5), `last_reviewed_by` ( `ForeignKey(User, null=True, on_delete=PROTECT, related_name='reviewed_husbandry')` , required when `published=True` ), `last_reviewed_at` (DateField, nullable; required when `published=True` ), `created_at` / `updated_at` (auto).

- **Meta** — ordering `('species__scientific_name',)` ; `verbose_name_plural = 'Species husbandry records'` .

### `HusbandrySource`

Child table for citations (BA §5 requires ≥1 source to publish).

- `husbandry` — `ForeignKey(SpeciesHusbandry, on_delete=CASCADE, related_name='sources')` .
- `label` — `CharField(max_length=500)` , required.
- `url` — `URLField(blank=True)` .
- `order` — `PositiveSmallIntegerField(default=0)` for stable admin ordering.
- Meta ordering `('order', 'id')` .

### Validation (model-level `clean` )

When `published=True` :

- At least one `HusbandrySource` row must exist (checked in `ModelAdmin.save_related` , since inlines aren't available at model `clean` time — raise `ValidationError` back to the change form).
- `last_reviewed_by` and `last_reviewed_at` must be non-null.

Not implicitly required: any of the value-field sections. A record can publish with just narrative + sources + reviewer — useful for sparse species.

## API

New DRF endpoint under the existing API app:

- `GET /api/species/{id}/husbandry/` — returns 200 with the full serialized `SpeciesHusbandry` (including nested `sources` , reviewer username + ORCID if present, `last_reviewed_at` , and a computed `review_is_stale` boolean set to `true` when `last_reviewed_at` is older than 24 months) **only if `published=True`** . Returns 404 otherwise — regardless of whether an unpublished draft exists. Public (no auth required). Cacheable (same revalidate window as species detail — 3600s).

- `GET /api/species/{id}/` — existing endpoint. Add a minimal field `has_husbandry` (boolean) at the top level, derived from `SpeciesHusbandry.objects.filter(species_id=id, published=True).exists()` . Do **not** embed the full husbandry payload here — the frontend fetches it lazily on the husbandry route. This is deliberate: the profile page stays fast and the husbandry payload stays cacheable independently.

Serializer exposes only the fields above. Never exposes `published` (implied by the 404 behavior) or internal timestamps beyond `last_reviewed_at` .

## Admin

- `SpeciesHusbandryAdmin` with inline `HusbandrySourceInline` (TabularInline, min zero, validated to ≥1 when `published=True` via `save_related` ).
- List display: species, published, last_reviewed_at, review_is_stale badge, source count.
- List filters: `published` , `sourcing_cares_registered_breeders` , `breeding_spawning_mode` .
- Search: `species__scientific_name` , `species__common_names__name` , `contributors` .

- Fieldsets grouped to match the template: Water / Tank / Diet / Behavior / Breeding / Difficulty Factors / Sourcing / Narrative / Governance.
- Help text on the Difficulty Factors fieldset: "Describe factors honestly. Do not attempt an overall difficulty label — the page deliberately surfaces factors, not a verdict."
- If `last_reviewed_at` is > 24 months old at edit time, render a warning row at the top of the change form: "Review is overdue; public page will show a 'review pending' note."
- Authoring permission: Tier 5 only at MVP. Tier 4 write access is post-MVP (tracked separately; matches `data/husbandry/TEMPLATE.md` workflow where Aleksei is the sole author pre-workshop).

## Acceptance Criteria

### AC-08.1 — Model shape matches template

**Given** a developer reads `data/husbandry/TEMPLATE.md` **When** they compare it to `SpeciesHusbandry` fields **Then** every non-metadata field in the template has a corresponding model field, with naming that is mechanically translatable (e.g. `water.ph_min` → `water_ph_min`).

### AC-08.2 — No aggregate difficulty column exists

**Given** the `SpeciesHusbandry` model **When** inspected **Then** there is **no** `difficulty` field; difficulty is represented exclusively by the seven `difficulty_*` factor fields.

### AC-08.3 — Publishing without a source is rejected

**Given** an admin creates a `SpeciesHusbandry` with no `HusbandrySource` inlines and sets `published=True` **When** they click Save **Then** the save is rejected with a form-level error "At least one source citation is required to publish." and the record is not marked published.

### AC-08.4 — Publishing without a reviewer is rejected

**Given** an admin creates a `SpeciesHusbandry` with at least one source but leaves `last_reviewed_by` or `last_reviewed_at` empty and sets `published=True` **When** they click Save **Then** the save is rejected with a field-level error naming the missing field(s), and the record is not marked published.

### AC-08.5 — Unpublished drafts are invisible to the public API

**Given** a `SpeciesHusbandry` exists for species X with `published=False` **When** an anonymous client requests `/api/species/{X}/husbandry/` **Then** the response is **404**, not 200 with a `published: false` flag.

**And Given** the same state **When** the client requests `/api/species/{X}/` **Then** the response body includes `"has_husbandry": false`.

### AC-08.6 — Published record is readable by Tier 1

**Given** species X has a published `SpeciesHusbandry` with one source and a reviewer **When** an anonymous client requests `/api/species/{X}/husbandry/` **Then** the response is 200 and includes all non-governance fields, the `sources` array, `last_reviewed_by` (username + ORCID if available), `last_reviewed_at`, `contributors`, and a computed `review_is_stale` boolean.

**And Given** the same state **When** the client requests `/api/species/{X}/` **Then** the response body includes `"has_husbandry": true`.

### AC-08.7 — Stale review is surfaced

**Given** a published `SpeciesHusbandry` with `last_reviewed_at` more than 24 months before today **When** the husbandry endpoint is requested **Then** `review_is_stale` is `true` in the response.

**And Given** the same record **When** an admin opens it in Django admin **Then** a warning banner "Review is overdue" renders at the top of the change form.

### AC-08.8 — One-to-one integrity

**Given** a `SpeciesHusbandry` exists for species X **When** an admin attempts to create a second `SpeciesHusbandry` for species X **Then** the save fails with a uniqueness error (enforced by the `OneToOneField`).

### AC-08.9 — Species deletion cascades

**Given** species X has a `SpeciesHusbandry` with sources **When** species X is deleted **Then** the `SpeciesHusbandry` row and its `HusbandrySource` rows are deleted (cascade).

### AC-08.10 — Unpublishing is reversible

**Given** a published `SpeciesHusbandry` **When** an admin toggles `published=False` and saves **Then** the save succeeds (no validation checks gate unpublishing), the public endpoint 404s, and `has_husbandry` flips to `false` on the species detail response.

## Out of Scope

- Frontend route `/species/[id]/husbandry/` and the teaser block on the profile (Gate 09).
- Contribute contact form (Gate 10).
- Photo/image attachments on husbandry records (post-MVP per BA §6).
- Version history / edit audit log (post-MVP; the `ConservationAssessment` audit pattern is not ported to husbandry in this gate).
- Tier 3+ write submission pipeline (post-MVP).
- Structured difficulty filter on the species directory (post-MVP per BA §Cross- Feature Impact; the factor fields exist but are not yet filter facets).
- FishBase "culture" field auto-pull (post-MVP).
- Public edit history, version diffs, comments, ratings (post-MVP).
- Richer `contributors` structure (free text at MVP is deliberate per BA §5).

## Dependencies

- `User` model (already in place).
- `Species` model (already in place).
- DRF wiring for new endpoints (existing species API app).
- Authoring workflow assumes Aleksei hand-enters 3–5 exemplars from `data/husbandry/TEMPLATE.md`. Q1 (authoring capacity) remains open; this gate is independent of final count.

## Sequencing / Deadline Note (June 1, 2026)

This gate is the **critical path** for the workshop demo. Target: merged by **2026-05-08** to leave two weeks for Aleksei to author exemplars against the live admin and for Gate 09 (frontend) to land on top with ~2

weeks of QA runway before June 1. If this gate slips past 2026-05-15, consider descoping Gate 10 (contribute form) to unblock Gates 08/09 for the demo.

## Test Writer Guidance

At this gate, the test writer should verify:

- Model-level: unique-per-species constraint, cascade-on-species-delete, `review_is_stale` boundary at exactly 24 months (on-day and off-by-one).
- Admin-level: publish-without-source rejection, publish-without-reviewer rejection, publish-with-both succeeds, unpublish has no validation gate.
- API-level: 404 vs 200 parity (unpublished → 404, published → 200); `has_husbandry` flag on species detail flips correctly on publish/unpublish/delete.
- **Adversarial:** attempting to POST/PUT/DELETE on the husbandry API as an anonymous or Tier 1-4 user should return 405 (read-only endpoint) or 401/403 depending on DRF config — never 200.
- **Adversarial:** a draft record for species Y where `published=False` but one source + reviewer are set must still 404 publicly. Do not leak drafts via ETag or any other side channel.
- **Adversarial:** flipping `published=False → True` without having added sources in the same transaction must not succeed.

## Risks and Open Questions

- **Authoring capacity (Q1, still open).** If Aleksei lands zero exemplars before Gate 09 frontend work begins, the frontend cannot be visually verified against real content. Mitigation: seed one fixture record from `TEMPLATE.md` (fully populated, marked `published=False`) as part of this gate's test fixtures so the frontend has at least one readable shape to develop against.
- **Reviewer ORCID surfacing. Resolved 2026-04-18:** the field exists — `accounts.User.orcid_id` (CharField, blank=True) at `backend/accounts/models.py:55`. Gate 08 serializer exposes it as `reviewer.orcid` (or `orcid_id` — pick one and match Gate 09's governance footer). No schema change needed; no follow-up required.
- **App location (`husbandry` vs folded into `species`). Resolved 2026-04-18:** new `husbandry` app. Architecture + BA both recommended new app (unanimous). Key reasons: consistent with existing per-bounded-context pattern; isolates migrations from the hot `species` table; natural home for upcoming contribute/moderation pipeline, photos, and version history; cost to move later (RunPython table renames) materially exceeds the half-day of scaffolding now. Constraint carried forward: **do not denormalize taxonomy or `iucn_status` onto `SpeciesHusbandry`** — one-to-one FK and read-through only.